Intelligent OCR for Scanned Documents — French / English / Arabic

Summary of the technical note: rationale for the OCR engine selection — July 2026

1. Problem

Scanned administrative forms, reports and handwritten material must be transcribed in **French, English and Arabic**, including mixed and right-to-left content. Classical OCR (Tesseract) detects *character shapes* with no understanding of context, layout or language, and degrades sharply on noisy scans, mixed scripts, tables and handwriting — precisely the cases at hand. A **Vision-Language Model** reads the image and produces text within a single network, and that joint understanding is what makes difficult documents tractable.

Three constraints shaped every decision: the reference test document is a **strictly confidential** safeguarding form containing names and medical details, so no document may leave the machine; the target hardware is a laptop with an **NVIDIA RTX 4050 (6 GB VRAM)**; and the solution must incur **no recurring cost**.

2. Solution

A **local-first** chain: ingestion (PDFs rasterised locally, one image per page) → preprocessing → VLM inference → Markdown structuring and language detection → optional quality control → delivery. Preprocessing runs on **OpenCV** (grayscale, non-local-means denoising, deskew, CLAHE contrast); binarisation is applied only on the Tesseract path, since the vision model reads contrast-enhanced grayscale better. PDFs are rasterised with **PyMuPDF** at 300 DPI — the resolution at which fine print stays legible — and read page by page.

Role	Model	Where it runs
Default engine	Qwen3-VL 4B instruct	Local, via Ollama, resident on the GPU
Second opinion (QA)	Qwen3-VL 8B instruct	Local, via Ollama
Classical baseline	Tesseract (`fra+eng+ara`)	Local, optional — benchmarking and hybrid mode
Optional cloud	Llama 4 Scout (vision)	Groq API

All four sit behind one abstraction, so engines can be swapped without touching the processing chain. Three recognition modes run over it: **VLM vision** (default), **hybrid** (Tesseract + VLM correction) and **Tesseract only** (baseline).

3. Key technical decisions

- **A VLM rather than classical OCR.** Classical OCR recognises glyphs; a VLM understands the page — context, layout and language. Tesseract is kept as an optional baseline so the claim is measured, not asserted.
- **Qwen3-VL.** Strong multilingual performance including Arabic, built for document tasks (dense text, tables, structured forms, Markdown output), open weights, no licence fee.
- **4B rather than 8B — measured, not assumed.** On the reference handwritten form the 4B scored **10/10 in 59 s** against **9/10 in 135 s** for the 8B: at 3.3 GB it stays fully resident in 6 GB of VRAM, while the 8B spills to the CPU. The smaller model is both faster and more complete on this hardware.
- **The `-instruct` tag is mandatory.** The base `qwen3-vl` tags run in *thinking* mode: they spend the whole token budget on internal reasoning and return an empty transcription.
- **Context window at 8192 tripled the speed.** Each page is a separate call, so the window only needs to hold one page. Raising it further wastes VRAM on the KV cache and forces model layers onto the CPU — about three times slower, for no accuracy gain.
- **Local (Ollama) as the default.** No document leaves the machine, no per-page fee, no rate limit, no internet needed — and fast enough on real hardware.

4. Quality control

On a dense form the dominant failure mode is **omission**, not hallucination: faint print, marginal notes, table cells and checkboxes get silently dropped, and simply reading the page twice does not help — the model skips the same things twice. Three independent tools address this from different angles, alongside CER/WER measurement where a ground truth exists.

Tool	Principle	Cost
Recall verification	A second pass receives the same image *plus its own draft*, under a prompt whose only job is to find what the draft missed. Handing the model its own output as a checklist turns open-ended transcription into a targeted search.	2× model calls
Two-model comparison	The page is read by two independent models and every disagreement is flagged. The 4B and 8B have *different* blind spots, so the diff surfaces real omissions a single pass would hide. Fully offline.	2.5–3×
Format validation	Postcodes, phone numbers and emails are format-constrained, so a share of errors is catchable **with no ground truth at all**: structural, consistency and length checks, with an unambiguous repair suggested when a single OCR confusion substitution fixes the token.	negligible

Format validation checks **form, not truth**: a well-formed but wrong value passes clean. A clean report means "no detectable format errors", never "no errors".

5. Delivery and performance

The OCR core is interface-agnostic. The primary deliverable is a **full-stack web platform** — React (Vite) single-page application, FastAPI backend with JWT authentication and PBKDF2 password hashing, MongoDB for a permanent searchable per-user document history — backed by a Streamlit prototype and a command-line interface that additionally exposes the three quality tools above.

Workload	Time	Notes
Printed page	25–29 s	RTX 4050, 300 DPI, `num_ctx` = 8192
Dense handwritten page	32–47 s	
Three-page printed PDF	86 s	measured
Ten-page form	about 5 minutes	
First page after idle	+10 s, one-time	model loads into VRAM

Rule of thumb: **time ≈ pages × 30 s**. Two parameters trade speed against completeness — the context window and the PDF rasterisation resolution — and both defaults deliberately favour completeness.

6. Known limitations

- Performance is tied to the presence of a GPU and its VRAM; without one, processing times become prohibitive.
- The figures come from one reference document on one machine: they indicate a trend, not a universal guarantee.
- Fidelity is not guaranteed at 100%. Given personal and medical data, **human proofreading remains necessary** — none of the three quality tools replaces it.
- Arabic handwriting remains the hardest case and accounts for most of the residual errors.
- The cloud path depends on a provider's model catalogue, which changes over time; the model identifier is therefore configurable.
- The format-validation rules are UK-specific and would need extending for other locales.

7. Conclusion

The system uses **Qwen3-VL-4B-Instruct** served locally via **Ollama** as its default OCR engine, chosen for its multilingual performance (notably Arabic), its handling of documents, tables and handwriting, and — critically — the fact that it runs **fully offline on a 6 GB laptop GPU**. The 4B size was selected over the 8B after direct measurement. A Groq / Llama 4 Scout cloud engine is retained behind the same abstraction as a higher-capacity alternative and as a benchmark. Around that engine the project delivers a complete chain — preprocessing, page-by-page PDF handling, structured Markdown output, three quality-control tools and a multi-user web platform — built to make human review short, not unnecessary.